AuctionHaus: Real-Time Shill-Bidding Detection via Online Bi

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Problem

Shill bidding inflates auction prices. Existing detectors operate **post-hoc** on completed logs. We need real-time detection to intervene **before** the auction closes.

Method	When?
Trevathan & Read (2007)	Post-auction
Ford et al. (2010)	Post-auction
Tsang et al. (2014)	Post-auction
This work	**Real-time**

System Architecture

- Node.js (TypeScript) · Express · Socket.io · PostgreSQL · Redis
- BullMQ workers (auto-bid ladder, notifications)
- In-process fraud engine — no external ML service
- Next.js frontend with live admin fraud dashboard

5-Feature Bid Graph

1. **Response time** (τ) — sub-500ms = bot
2. **Bid frequency** (ϕ) — bids/min in 30-min window
3. **Increment ratio** (ρ) — scripted min-increment pattern
4. **Seller co-occurrence** (σ) — same seller, many auctions
5. **Reciprocity** (γ) — mutual outbidding = collusion ring

Sliding window graph updated on each bid event ($<$ 1ms overhead). Features z-scored, scored by logistic regression:

$$P(\text{shill}|\mathbf{z}) = \sigma(\beta_0 + \sum_i \beta_i z_i)$$

Results

	Baseline	**This work**
Precision	0.000	**0.864**
Recall	0.000	**1.000**
F1	0.000	**0.927**
AUC	0.620	**0.890**

Evaluated on synthetic simulator runs (4 agent personas).

W1: Commit-Reveal Sealed Bids

Client commits H = SHA-256(amountHex‖:‖nonce) during live phase. Server stores hash only. Reveal after close.

- **Hiding**: server cannot learn amount during bidding
- **Binding**: bidder cannot change amount after committing

W2: Redis Stream Sequencer

Concurrency	Baseline p95	**Stream p95**
10	1,284ms	**48ms**
100	5,772ms	**8.1ms**

32.5× throughput scaling at 100 concurrent bidders.

Conclusion

AuctionHaus is the first auction platform with **in-auction, real-time** shill detection. F1 improves from 0.000 (outbid-count baseline) to 0.927 (5-feature LR). Cryptographic sealed bids and Redis Stream sequencer add production-grade hardening and research novelty beyond standard CRUD.

Code: https://github.com/prathamreet/auctionhaus